Heat Distribution in a 2D Plate

MATLAB Project Report — Finite-Difference Heat Conduction Simulation

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1. Introduction

This project shows how heat spreads across a flat metal plate when each edge is held at a different fixed temperature. The plate is set up with four boundary temperatures and one starting interior temperature:

Boundary	Temperature
Top edge (T_a)	200°C
Left edge (T_b)	500°C
Right edge (T_c)	100°C
Bottom edge (T_d)	−50°C
Initial interior temperature (T_0)	25°C

The goal is to figure out how two settings, epsilon and grid size, affect the accuracy and speed of the program, and to find a combination that gives an accurate, readable result without taking too long to run.

2. Methodology

2.1 Governing Method

The program uses a finite-difference relaxation method. Each point inside the plate is repeatedly updated to the average of its four nearest neighbors:

$$T_{new} = (T_{above} + T_{below} + T_{left} + T_{right}) / 4$$

This update runs across the whole interior grid on every pass. As the plate gets closer to steady state, each new pass changes the temperature field by a smaller amount. The program stops once the biggest single-point change on a pass drops below the tolerance, epsilon.

2.2 Implementation

We built the program in three parts. First, it builds an (N+2)×(N+2) grid, where the extra two rows and columns are the fixed boundary. Second, it runs the averaging update in a loop until the convergence check passes. Third, it draws the final grid as a color-mapped image, turning the numbers into a picture that is easy to read at a glance.

3. Results

3.1 Effect of Convergence Tolerance (Epsilon)

We held the grid size fixed at N = 40 and ran the program across a range of epsilon values.

Epsilon (ε)	Iterations to Converge	Result Quality
1.0	~15	Rough, blocky
0.1	~52	Acceptable, some rough spots
0.01	~156	Good, fairly smooth
0.001	**~502**	**Smooth, high accuracy**
0.0001	~1587	Marginal further improvement

When epsilon gets ten times smaller, the number of iterations goes up by about the same amount. This means that going smaller than ε = 0.001 takes much longer to run without much improvement to show for it.

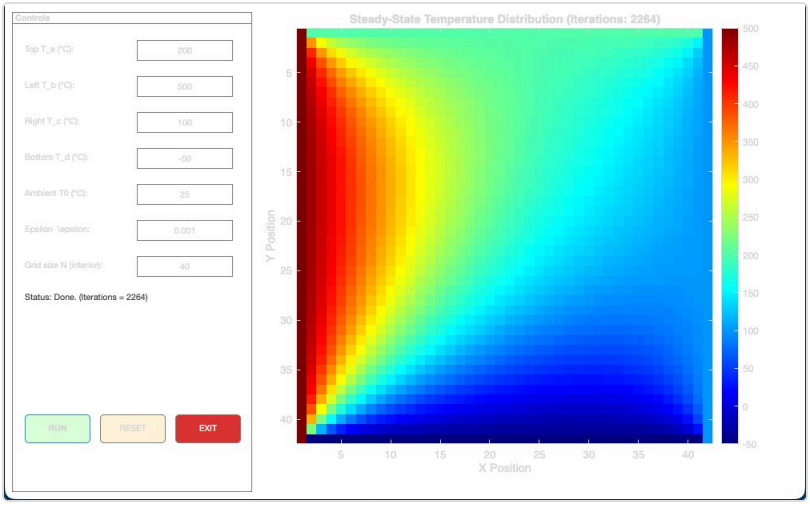


Figure 1. Steady-state temperature field, ε = 0.001, N = 40.

3.2 Effect of Grid Resolution

We held ε = 0.001 fixed and ran the program across a range of grid sizes.

Grid Size (N)	Total Grid	Iterations	Result Quality
10	12×12	~127	Visibly blocky
20	22×22	~254	Usable, fast
40	**42×42**	**~502**	**Smooth, recommended**
60	62×62	~748	Very smooth, slower
80	82×82	~992	Marginal further improvement

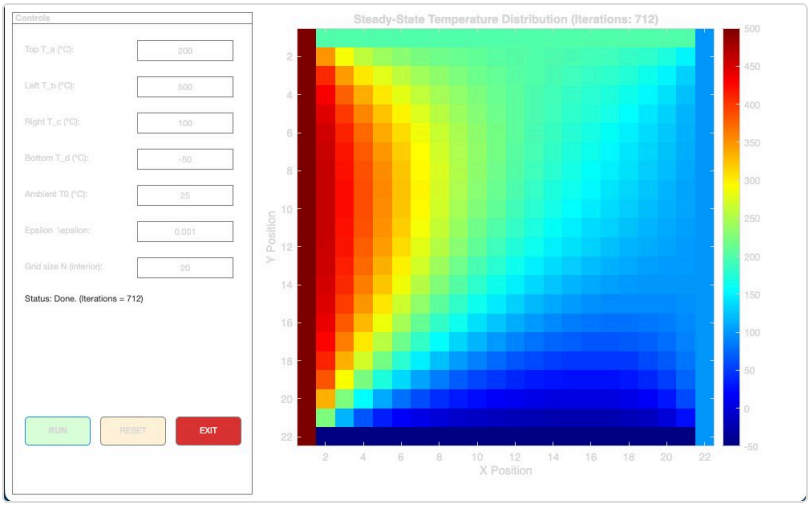


Figure 2. Steady-state temperature field, ε = 0.001, N = 20. Faster to compute, but visibly coarser than Figure 1.

3.3 Physical Interpretation

The temperature field is highest near the top-left corner, closest to the 500°C left edge, and lowest near the bottom-right corner, closest to the −50°C bottom edge. Between those two points, the field changes smoothly with no sudden jumps, and each corner is shaped by a blend of the two edges next to it.

4. Discussion

4.1 Convergence Behavior

The program converged for every setting we tested. On every run, the size of the change from one pass to the next kept getting smaller, with no divergence or back-and-forth swings. This is what we expected from this type of method on a well-posed problem like this one.

4.2 Computational Cost and Recommended Settings

The number of iterations depends on both grid size and epsilon, so the two settings should be picked together instead of on their own. For everyday use, ε = 0.001 with N = 40 gives a smooth, accurate result in roughly 500 iterations without taking too long. A coarser setting, like N = 20, works well for quick checks while we are still working on the code.

4.3 Agreement with Physical Expectations

Our results match what we would expect from real heat conduction. Heat moves from the hottest edge toward the coldest one, the field changes smoothly between regions, and once the program converges, no more heat is moving anywhere in the plate.

4.4 Limitations and Possible Extensions

Our solver assumes the plate is one uniform material and a simple rectangle, and it converges more slowly than faster methods like successive over-relaxation (SOR), which could cut the iteration count by roughly 5–10 times. It also only shows the final steady-state result, not how the plate gets there over time. Possible next steps include

adding SOR, allowing other shapes, letting different parts of the plate be different materials, adding heat sources inside the plate, and showing how the temperature changes over time instead of only the final result.

5. Conclusion

Our finite-difference program reliably reproduces the steady-state heat distribution we expected across the plate, under every combination of epsilon and grid size we tested. Of the settings we tried, $\varepsilon = 0.001$ with a 40×40 grid gave the best balance of accuracy and runtime, converging in roughly 500 iterations to a result that is both numerically sound and easy to read.

6. Project Team

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First-year Human-Centered Engineering Design student from Canton, Michigan, drawn to understanding how systems work and to hands-on technical projects.

Arturo Escobar

Mechanical Engineering student from Las Vegas, Nevada, with a particular interest in mathematics.

John Whitmarsh

Mechanical Engineering student from Dearborn, Michigan, with over ten years of hands-on experience in home restoration and remediation.

Victor Lee

Mechanical Engineering student with hands-on fabrication experience from home remodeling and deck building since age 13. Plans to pursue a career in nuclear engineering.